

**MIT AI2 204**

# **IoT with MIT App Inventor**

**Fundamental**

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# **IoT with MIT App Inventor**

**Start 7:05PM**

**Please update your name with studentID  
In Zoom meeting**

# Sending Temperature and Humidity To your android device

Description: In this project, a temperature and humidity sensor module is connected to your Raspberry Pi. The Raspberry Pi sends the ambient temperature and humidity measurements every 10 seconds to the Android mobile phone where they are displayed on the screen.



# Sending Temperature and Humidity To your android device

Hygrothermograph is an important tool in our lives to give us data on the temperature and humidity in our environment. In this project, we will use the RPi to read Temperature and Humidity data of the DHT11 Module.

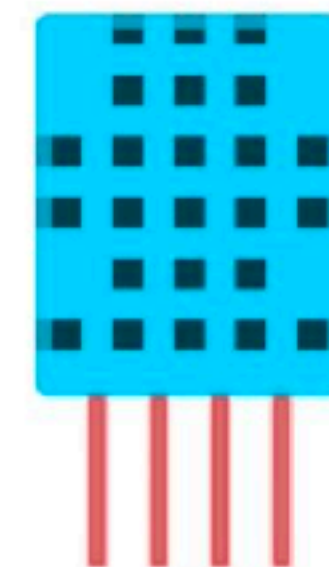
## Component List

Raspberry Pi (with 40 GPIO) x1  
GPIO Expansion Board & Ribbon Cable x1  
Breadboard x1

Jumper Wire x4



DHT11 x1

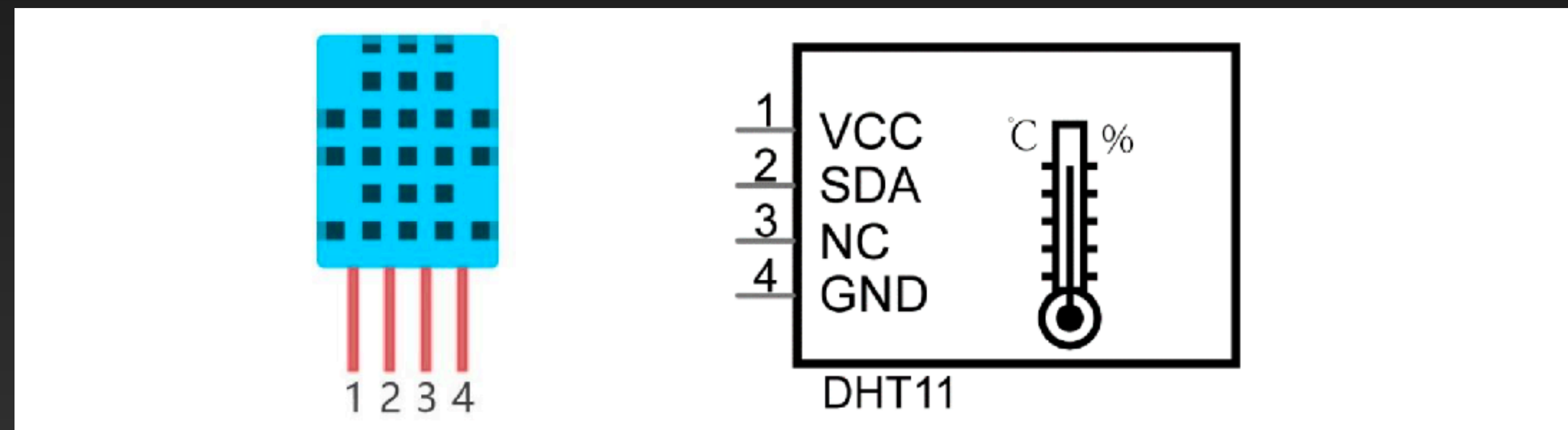


Resistor 10kΩ x1



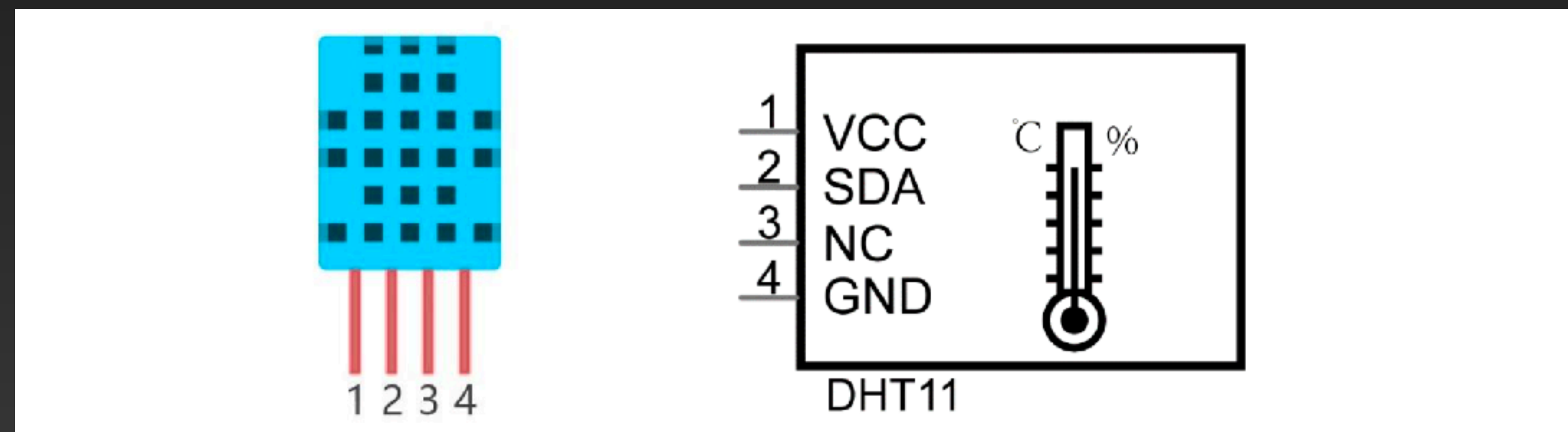
# Sending Temperature and Humidity To your android device

The Temperature & Humidity Sensor DHT11 is a compound temperature & humidity sensor, and the output digital signal has been calibrated by its manufacturer.



# Sending Temperature and Humidity To your android device

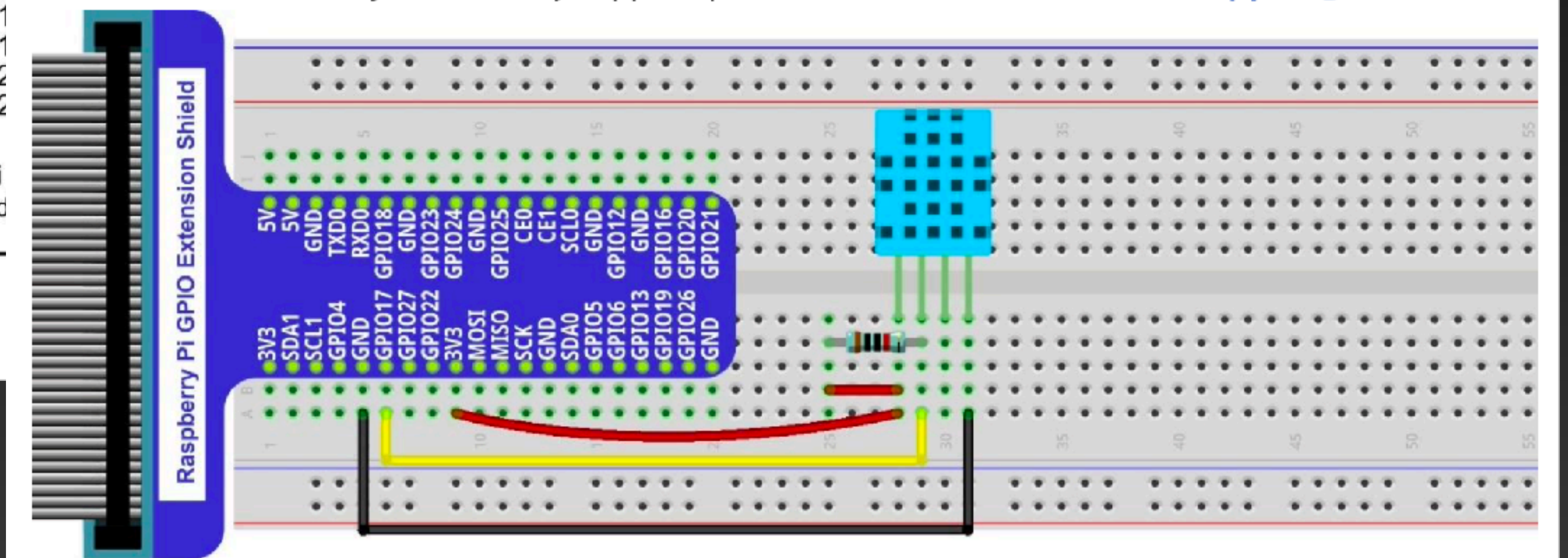
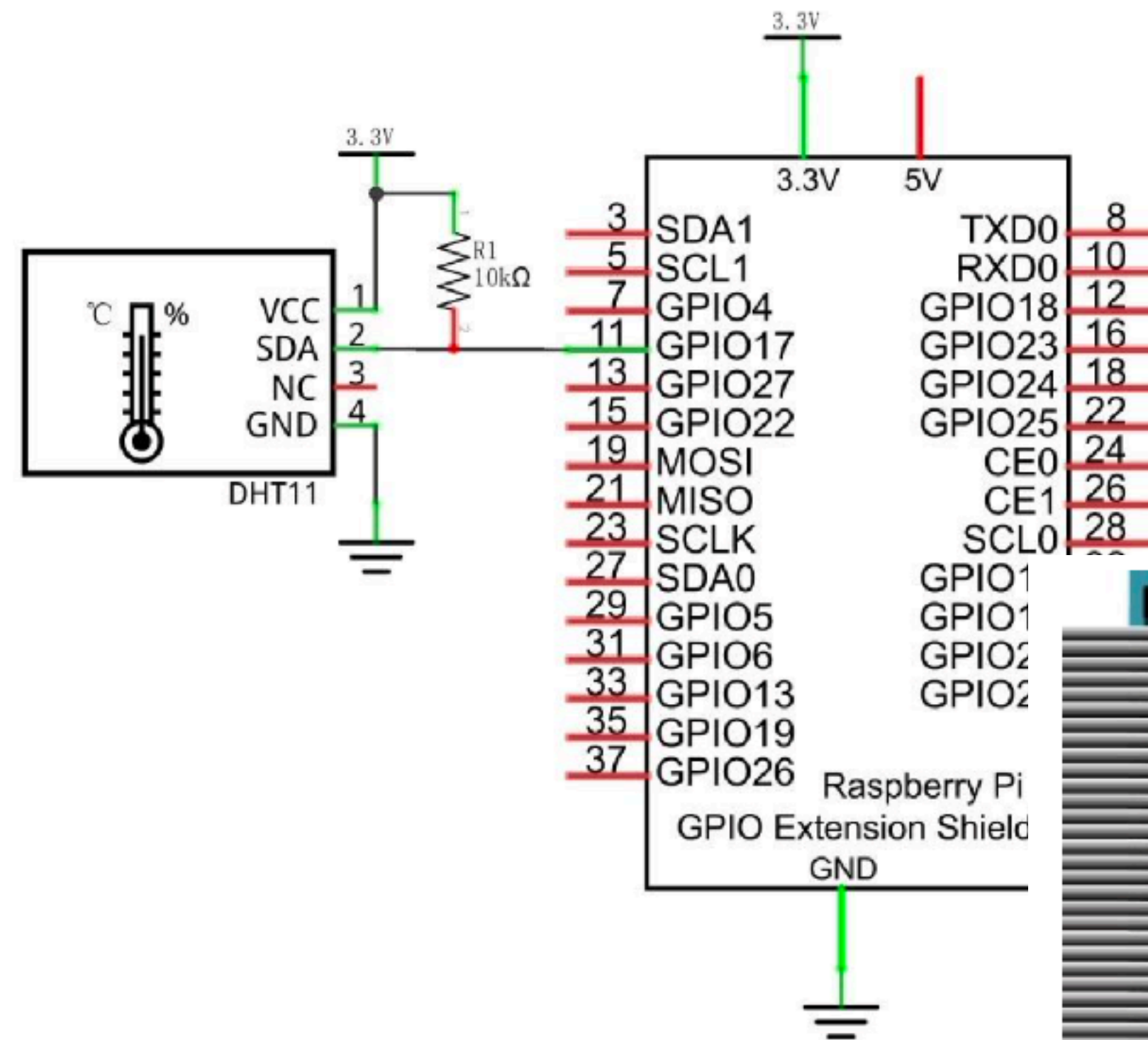
After being powered up, it will initialize in 1 second. Its operating voltage is within the range of 3.3V-5.5V. The SDA pin is a data pin, which is used to communicate with other devices. The NC pin (Not Connected Pin) are a type of pin found on various integrated circuit packages. Those pins have no functional purpose to the outside circuit (but may have an unknown functionality during manufacture and test). Those pins **should not be connected** to any of the circuit connections.





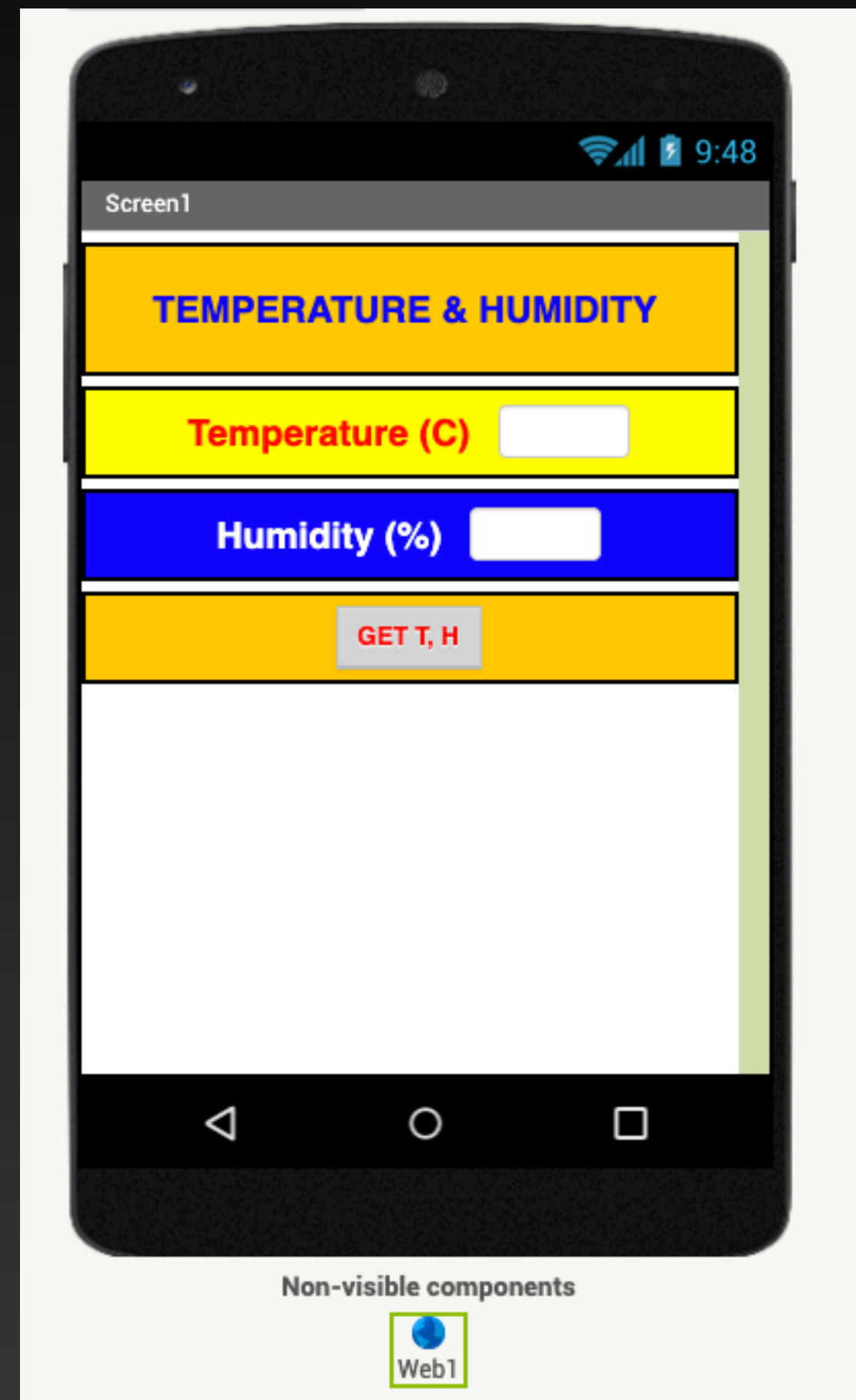
# Sending Temperature and Humidity To your android device

Schematic diagram



# Sending Temperature and Humidity To your android device

Build your MIT App



```
initialize global TH to ""
initialize global RaspberryPi to "http://192.168.1.xxx"

when Button1.Click
do
  set Web1.Url to get global RaspberryPi
  call Web1.Get

when Web1.GotText
  url responseCode responseType responseContent
do
  set global TH to get responseContent
  set TextBox1.Text to segment text get global TH
    start 1
    length 2
  set TextBox2.Text to segment text get global TH
    start 4
    length 2
```



# Sending Temperature and Humidity To your android device

Prepare the environment:(pi4)

```
sudo pip3 install adafruit-circuitpython-dht
```

There is new updates to the library:

[https://github.com/adafruit/Adafruit\\_CircuitPython\\_DHT](https://github.com/adafruit/Adafruit_CircuitPython_DHT)

# Sending Temperature and Humidity To your android device

Prepare the environment:(pi5)

```
mkdir dht_test
```

```
cd dht_test
```

```
python3 -m venv myenv
```

```
source myenv/bin/activate
```

```
python3 -m pip install adafruit-circuitpython-dht
```

```
python3 -m pip install flask
```

——-Find the python3 location and run with sudo command——-

```
sudo /home/pi/DHT11/myenv/bin/python3 DHT11Test.py
```

# Sending Temperature and Humidity To your android device

Write the python's  
control program

```
9 import time
10 from flask import Flask, render_template
11 import RPi.GPIO as GPIO
12 import adafruit_dht
13 from board import *
14
15 # GPIO17
16 SENSOR_PIN = D17
17
18 app = Flask(__name__)
19
20 @app.route('/', methods=['GET', 'POST'])
21 def get_data():
22
23     dht11 = adafruit_dht.DHT11(SENSOR_PIN, use_pulseio=False)
24     temp = dht11.temperature
25     hum = dht11.humidity
26     tempint = int(temp)
27     humint = int(hum)
28     if tempint < 10:
29         datat = "0" + str(tempint)
30     else:
31         datat = str(tempint)
32     datath = datat + "," + str(humint)
33     return(datath)
34
35 if __name__ == '__main__':
36     app.run(debug=True, port=80, host='0.0.0.0', use_reloader=False)
37
38
```



## Prepare the Env and Test the Sensor

```
git clone --depth 1 https://github.com/freenove/  
Freenove\_Ultimate\_Starter\_Kit\_for\_Raspberry\_Pi
```

```
mv Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi ~/Freenove_Kit/
```

```
cd Freenove_Kit/Code/Python_GPIOZero_Code/21.1.1_DHT11/
```

```
python DHT11.py
```

# Sending Temperature and Humidity To your android device

```
pi@pi5: ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Libs/Python-Libs $ ls
ADCDevice-1.0.2.tar.gz  ADCDevice-1.0.3.tar.gz  Freenove_DHT11  spidev-3.6.tar.gz
pi@pi5:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Libs/Python-Libs/Freenove_DHT11 $ sudo python setup.py
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian-security bookworm-security InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Hit:4 http://archive.raspberrypi.com/debian bookworm InRelease
Reading package lists... Done
```

```
pi@pi5: ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_GPIOZero_Code $ ls
00.0.0_Hello          06.2.1_Alertor          14.1.1_Relay          21.1.1_DHT11
01.1.1_Blink          07.1.1_ADC             15.1.1_Sweep          22.1.1_MatrixKeypad
02.1.1_ButtonLED      08.1.1_Softlight       16.1.1_SteppingMotor  23.1.1_SenseLED
02.2.1_Tablelamp      09.1.1_ColorfulSoftlight 17.1.1_LightWater02   24.1.1_UltrasonicRanging
03.1.1_LightWater      10.1.1_Nightlamp       18.1.1_SevenSegmentDisplay 25.1.1_MPU6050
04.1.1_BreathingLED   11.1.1_Thermometer     18.2.1_StopWatch      26.1.1_WebIO
05.1.1_ColorfulLED    12.1.1_Joystick        19.1.1_LEDMatrix      27.2.1_LightWater03
06.1.1_Doorbell       13.1.1_Motor           20.1.1_I2CLCD1602
pi@pi5:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_GPIOZero_Code $ cd 21.1.1_DHT11
pi@pi5:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_GPIOZero_Code/21.1.1_DHT11 $ python DHT11.py
Program is starting ...
Measurement counts: 1
```



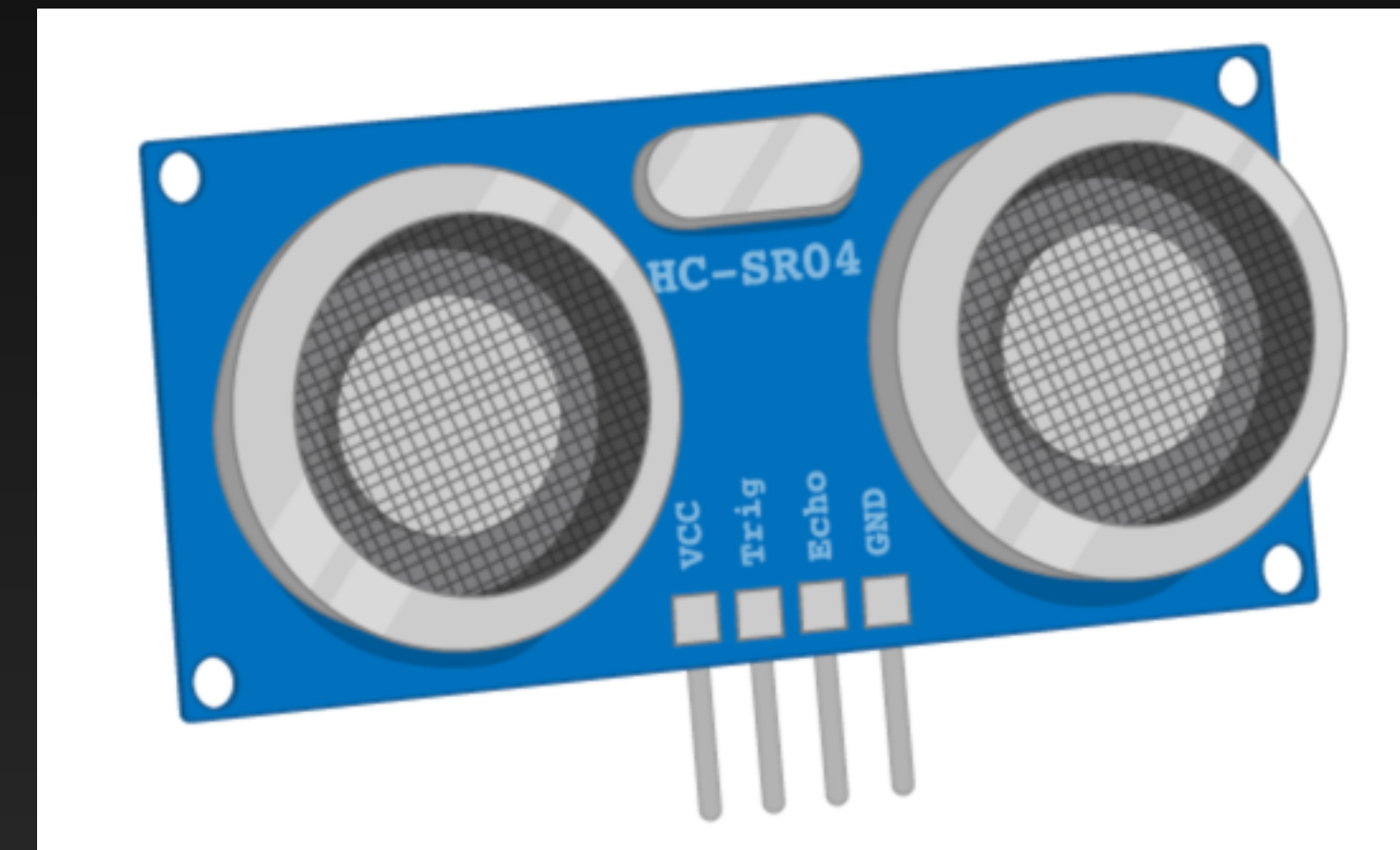
## Setup Distance Sensor

An ultrasonic distance sensor is a device that sends out pulses of ultrasonic sound, and measures the time they take to bounce off nearby objects and be reflected back. They can measure distances fairly accurately (up to about a meter).

# Setup Distance Sensor

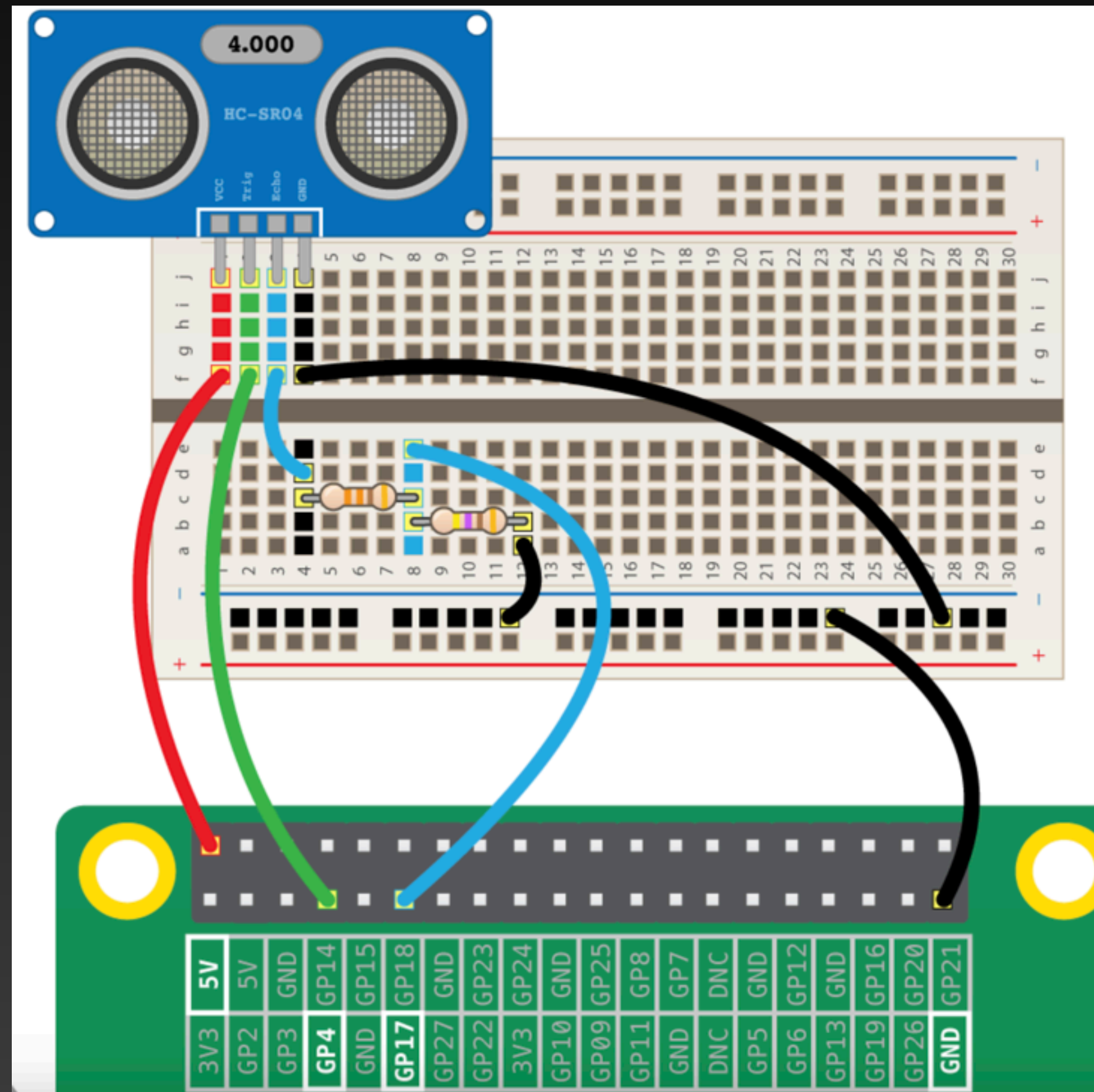
An ultrasonic distance sensor has four pins. They are called Ground (Gnd), Trigger (Trig), Echo (Echo), and Power (Vcc).

To use an ultrasonic distance sensor, you need to connect the Gnd pin to the ground pin on the Raspberry Pi, the Trig and Echo pins to GPIO pins on the Raspberry Pi, and the Vcc pin to the 3V3 pin on the Raspberry Pi.



# Setup Distance Sensor

The circuit connects to two GPIO pins (one for echo, one for trigger), the ground pin, and a 5V pin. You'll need to use a pair of resistors ( $220\Omega$ ) as a potential divider:





## Setup Distance Sensor

```
from gpiozero import DistanceSensor
ultrasonic = DistanceSensor(echo=17, trigger=4)
while True:
    print(ultrasonic.distance)
```

# Homework

Finish class project to setup Temperature and Humidity sensor, and post questions you have encounter